



DESN2000 SENG

Part 2 Group Report

LaTeX Framework

Project title	<i>[Insert project title]</i>
Team members	<i>[Insert team member names and zIDs]</i>
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1 Introduction and Part 2 Scope

Part 1 used surveys and interviews with four stakeholder groups to define the problem and develop personas. This report applies the Ideate stage to that evidence. It compares three technical concepts, selects one through a weighted decision matrix, and translates the selected direction into a Jira backlog and a preliminary prototype.

The report focuses on the student journey from accommodation search to an agreed shared living arrangement. Property manager stories remain in the backlog as secondary scope.

2 Problem Statement and Research Justification

2.1 Problem Statement

The following statement was approved in Part 1 and is reproduced without alteration.

Create a mobile platform which allows university students and shared accommodation residents to seamlessly manage household responsibilities, expenses and housemate communication, while also enabling property managers to streamline tenancy administration and tenant communication, to reduce conflict between tenants, streamline property management, and improve the overall shared living experience.

The statement covers student and property manager needs. This report narrows the preliminary prototype to the student path. A student searches for an option, checks verification, compares compatibility, starts contact in the app, and acknowledges a household rule. Property manager functions remain outside Sprint 1.

2.2 Research Justification

Table 1 links the strongest Part 1 evidence to a user need and a backlog response. This traceability keeps the proposed features grounded in the research.

Table 1. Research traceability.

Research evidence	User need	Backlog response
All 16 international respondents reported concern about rental scams. A further 43.75% had encountered misleading or false listing information.	The international student persona needs clear evidence about a profile or listing before making contact.	Show profile verification status in US-11 and clear listing details in US-13.
87.5% rated housemate compatibility as very or extremely important. A further 68.75% had experienced conflict with a previous housemate.	The long term housemate persona needs to identify practical mismatches before committing.	Compare living preferences in US-12.
44% identified communication difficulty as the hardest part of the search. More than 80% were willing to use preference based matching.	The international student persona needs first contact without early disclosure of personal details.	Keep first contact inside the app through US-09.
All surveyed domestic students relied on verbal household agreements and had experienced conflict over responsibilities. A further 66.67% wanted better chore tracking.	A new share house resident needs accepted expectations to remain visible after move in.	Store acknowledged household rules through US-02 and US-10.

3 Solution Ideation

3.1 Research to Ideation Transition

The evidence in Table 1 was reframed as four design needs. Table 2 sets the boundary for concept generation and later evaluation.

Table 2. Research issues and design needs.

Research issue	Design need
Scam concern and misleading information	Visible verification status and clear listing data
Conflict caused by poor living fit	Structured preference comparison
Risk during first contact	In app messaging before contact disclosure
Verbal household expectations	Recorded rules with acknowledgement

The resulting design question was as follows.

How might we help students move from an uncertain accommodation search to an informed shared living agreement?

Each concept was assessed against this question.

3.2 Generated Technical Concepts

The team generated three concepts with different technical boundaries. Table 3 gives the comparison. Tables 4 to 6 describe how each concept operates.

Table 3. Overview of the generated concepts.

Concept	Name	Main user task	Technical boundary
A	Verified Contact Gateway	Check status before first contact	Verification state and protected messaging
B	Explainable Compatibility Matcher	Compare practical living fit	Preference model and explainable score
C	Trust to Agreement Journey	Move from search to an acknowledged rule	Integrated flow across the core services

Table 4. Concept A. Verified Contact Gateway.

Aspect	Description
Purpose	Reduce uncertainty during first contact with an unfamiliar person.
Operation	The system records each verification check as <i>unverified</i> , <i>pending</i> , <i>verified</i> , <i>rejected</i> or <i>expired</i> . The interface shows the status and hides submitted evidence. Messaging does not expose personal contact details.
Research fit	Responds to scam concern and reluctance to contact unknown people.
Limitation	A verified identity does not establish compatibility or future conduct.

Table 5. Concept B. Explainable Compatibility Matcher.

Aspect	Description
Purpose	Help users identify practical mismatches before committing.
Operation	Users record living preferences and mark mandatory constraints. The matcher removes candidates who breach a mandatory constraint. It then calculates a weighted score and shows the factors that affected the result.
Research fit	Responds to the importance of daily living compatibility in the Part 1 findings.
Limitation	The score supports discussion but cannot predict a successful household.

Table 6. Concept C. Trust to Agreement Journey.

Aspect	Description
Purpose	Connect the accommodation decision with an explicit household agreement.
Operation	The flow links profile setup, status display, compatibility comparison, in app contact, and rule acknowledgement. Each step passes only the data needed by the next step. Later household tools remain outside the first release.
Research fit	Connects uncertainty before move in with unclear expectations after move in.
Limitation	Its wider scope raises delivery and privacy risk.

Concepts A and B isolate one decision in the user journey. Concept C integrates those decisions and adds agreement. All three remained candidates for the weighted evaluation.

3.3 Ideation Outcome

The ideation stage produced three candidates with different value and delivery risk. No concept was selected at this stage. All three entered the weighted decision matrix in Section 4.

3.4 Ideation Method

Required content

Name and explain the ideation method used by the team. Show how the method started from the problem definition and produced multiple ideas. Include evidence from the actual ideation activity.

Table 7. Ideation method template.

Ideation stage	Method used	Team activity	Output
[Stage 1]	[Method]	[What the team did]	[Produced artefact]
[Stage 2]	[Method]	[What the team did]	[Produced artefact]
[Stage 3]	[Method]	[What the team did]	[Produced artefact]

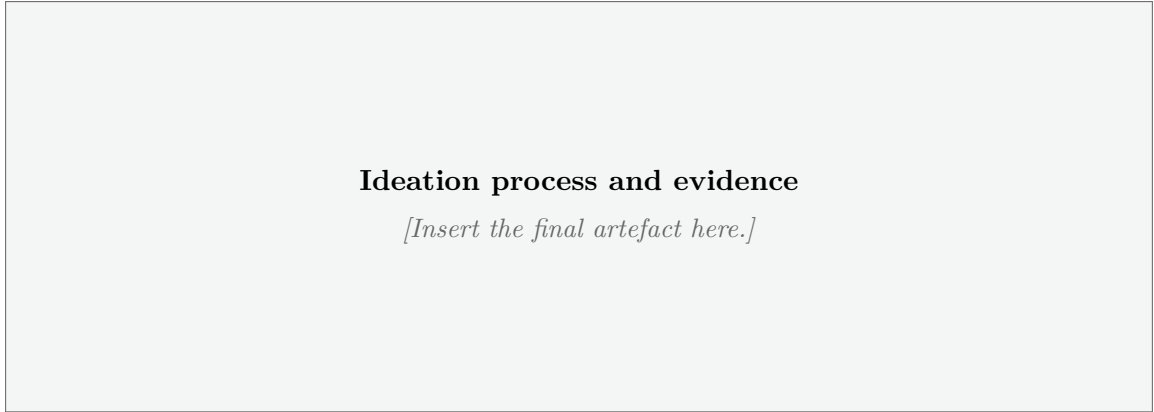


Figure 1. Ideation process and evidence.

4 Concept Selection and Prioritisation

4.1 Selection Method

The weighted decision matrix selected the concept. MoSCoW prioritisation then set the delivery scope. This separation prevented a broad concept from being treated as a commitment to deliver every related feature.

Each concept was rated from 1 to 5. A rating of 1 indicated poor fit. A rating of 3 indicated partial fit with material limits. A rating of 5 indicated full fit under the stated prototype assumptions. The weighted total was calculated as

$$S_j = \sum_i w_i r_{ij}$$

where w_i is the criterion weight and r_{ij} is the rating for concept j .

Table 8. Selection criteria and weights.

Criterion	Weight	Rationale
Research evidence fit	30%	Measures direct traceability to the Part 1 findings and personas.
User risk reduction	20%	Measures how much evidenced user risk the concept reduces.
Feasibility within ten weeks	20%	Measures whether the team can prototype and revise the concept within the schedule.
Prototype testability	15%	Measures whether users can complete observable tasks in a clickable flow.
Innovation beyond isolated tools	10%	Measures whether the concept creates a research grounded journey rather than copying one existing function.
Accessibility and privacy risk	5%	Measures whether foreseeable risks can be controlled. An unacceptable risk remains a rejection condition.

The final ratings in Table 9 must match the scoring evidence in Figure 2.



Figure 2. Week 6 concept-selection evidence: independent ratings and consensus discussion.

4.2 Decision Matrix

The weighted total is the sum of each rating multiplied by its weight. The maximum score is 5.00.

Table 9. Weighted decision matrix.

Selection criterion	Weight	Concept A	Concept B	Concept C
Research evidence fit	30%	4	4	5
User risk reduction	20%	4	4	5
Feasibility within ten weeks	20%	5	4	3
Prototype testability	15%	4	5	4
Innovation beyond isolated tools	10%	3	4	5
Accessibility and privacy risk	5%	4	4	3
Weighted total	100%	4.10	4.15	4.35

Concept C ranks first with 4.35. It fits the research and covers more of the target journey. Its lower feasibility and privacy rating exposes the main delivery risk.

4.3 Selected Concept and Rationale

Concept C was selected because it links the strongest research findings in one user path. Concept A stops after safer first contact. Concept B stops after the match decision. Concept C continues to an acknowledged household rule.

The preliminary prototype uses a narrow release slice to control scope. Table 10 aligns the priority of every story with the Sprint 1 path shown later in the report.

Table 10. MoSCoW priorities for preliminary prototype delivery.

Priority	User stories	Delivery decision
Must have in Sprint 1	US-02, US-09, US-10, US-11, US-12, US-13	These stories form one testable path from search to an acknowledged rule.
Should have later	US-01, US-03, US-04, US-05, US-06, US-14	These stories extend the agreement into routine household follow through.
Could have	US-07, US-08, US-15	These stories add supporting communication or reputation value without blocking the core path.
Will not have in the preliminary prototype	US-16, US-17, US-18	These stories introduce a separate role and additional permissions. They remain in the product backlog.

Only the Must have row defines Sprint 1. The same release slices are used in the backlog tables and must match Jira.

5 Product Backlog Structure

5.1 Epic Overview

The backlog contains four Epics. Each Epic groups stories that deliver a distinct user outcome. Stable story identifiers preserve links between this report, Jira, acceptance criteria, and prototype evidence.

Table 11. Epic structure of the product backlog.

Epic ID	Epic name	Epic goal	Linked user stories
EP-01	Housemate Discovery and Trust	Help an accommodation seeker assess an option before commitment.	US-09, US-11, US-12, US-13
EP-02	Shared Household Living	Make household decisions visible after move in.	US-01 to US-08 and US-10
EP-03	Accountability Extensions	Hold deferred reminders and review functions.	US-14, US-15
EP-04	Property Management	Support listing publication and tenant communication.	US-16, US-17, US-18

5.2 User Stories

Each story uses the standard user, goal, and value structure. The scope column follows Table 10.

Table 12. EP-01 Housemate Discovery and Trust user stories.

Story ID	User story	Scope	Jira ticket
US-09	As an international student, I want to message a potential housemate in the app before sharing contact details, so that I can make first contact with less exposure to scams or harassment.	Sprint 1	ST4-35
US-11	As an international student, I want to see a profile's verification status, so that I can assess the identity signal before contacting that person.	Sprint 1	ST4-38
US-12	As an accommodation seeker with an irregular schedule, I want to compare potential housemates against my living preferences, so that I can identify practical mismatches before committing.	Sprint 1	ST4-39
US-13	As a student moving out of home, I want to filter listings by location and cost, so that I can narrow the search to suitable options.	Sprint 1	ST4-43

Table 13. EP-02 Shared Household Living user stories.

Story ID	User story	Scope	Jira ticket
US-01	As a new share house resident, I want to assign rotating chores and view completion history, so that each contribution remains visible.	Later sprint	ST4-23
US-02	As an international student in a new household, I want to view the accepted house rules in one place, so that I can understand what the household expects.	Sprint 1	ST4-24
US-03	As a new share house resident, I want housemates to review a proposed chore system before it becomes active, so that the roster reflects a shared decision.	Later sprint	ST4-26
US-04	As a share house resident, I want to record a shared expense and who paid it, so that the cost can be divided among the relevant housemates.	Later sprint	ST4-28
US-05	As a share house resident, I want to view outstanding balances, so that I know who owes money and how much.	Later sprint	ST4-29
US-06	As an international student, I want to divide a utility bill among selected housemates, so that my share is clear.	Later sprint	ST4-32
US-07	As a resident who is often away, I want to post household notices in a dedicated space, so that important updates are not lost in chat.	Backlog	ST4-33
US-08	As a resident, I want to raise a household issue without showing my identity to other residents, so that I can report a sensitive concern with less social pressure.	Backlog	ST4-34
US-10	As a resident with different noise habits from my housemates, I want to propose a house rule and record each acknowledgement, so that the active rule is clear during a dispute.	Sprint 1	ST4-36

Table 14. EP-03 Accountability Extensions user stories.

Story ID	User story	Scope	Jira ticket
US-14	As a resident, I want an in app reminder when my chore or payment is overdue, so that I can act before a housemate must follow up.	Later sprint	ST4-25
US-15	As an accommodation seeker, I want to read reviews from confirmed former housemates, so that I can consider past shared living experience before choosing a match.	Backlog	ST4-41

Table 15. EP-04 Property Management user stories.

Story ID	User story	Scope	Jira ticket
US-16	As a property manager, I want to record rent status and send overdue reminders, so that I can follow up late payments efficiently.	Secondary	ST4-30
US-17	As a property manager, I want to publish an accurate rental listing, so that students can assess the property and enquire through the platform.	Secondary	ST4-40
US-18	As a property manager, I want to send an update to every current tenant at a property, so that important information reaches the correct audience.	Secondary	ST4-37

6 Acceptance Criteria

The acceptance criteria are stored in the matching Jira tickets and reproduced in the included file. Each story must include testable Given, When and Then statements for the main path. Relevant validation failures and boundary cases must also be covered. The report and Jira wording must match.

6.1 EP-01 Housemate Discovery and Trust

Table 16. Acceptance criteria for US-09 (Jira ST4-35).

Acceptance ID	Scenario	Given	When	Then
AC-US09-01	Success	two users are connected on the platform	they open a conversation	they can send and receive messages through a secure in-app system without either party's personal contact details being visible
AC-US09-02	Success	both users mutually agree to share contact details	they both opt in	the system exchanges their contact details between them
AC-US09-03	Success	a user feels unsafe	they block or report another user from the messaging interface	the reported user is flagged for review and the conversation is ended
AC-US09-04	Error	a user attempts to include personal contact details in a message	the system detects this	it warns the user and recommends using the opt-in contact sharing feature instead
AC-US09-05	Error	a user attempts to contact someone who has blocked them	they try to send a message	the system prevents the message from being sent without revealing that they have been blocked
AC-US09-06	Edge case	a user's account is flagged as unverified	they attempt to initiate a conversation	the system displays a warning to the recipient that the sender is unverified before they choose to respond

Table 17. Acceptance criteria for US-11 (Jira ST4-38).

Acceptance ID	Scenario	Given	When	Then
AC-US11-01	Success	a user has submitted a valid form of identification	the verification is approved	their profile displays a verified identity badge visible to all other users
AC-US11-02	Success	a user is searching for housemates	they apply the verified only filter	only profiles with a verified identity badge appear in search results
AC-US11-03	Success	a user views an unverified profile	they open it	a visible warning indicator is displayed clearly at the top of the profile
AC-US11-04	Error	a user submits an expired or unreadable identification document	the system processes it	the verification is rejected and the user is notified with instructions to resubmit a valid document
AC-US11-05	Error	a user attempts to access the platform's core matching features without completing verification	they try to proceed	the system prompts them to complete verification first
AC-US11-06	Edge case	a verified user's identification document expires	the system detects the expiry	it notifies the user to resubmit updated documentation to maintain their verified status

Table 18. Acceptance criteria for US-12 (Jira ST4-39).

Acceptance ID	Scenario	Given	When	Then
AC-US12-01	Success	a user has completed their lifestyle preference profile	they search for housemates	results are automatically ranked by a compatibility score calculated from matching preferences
AC-US12-02	Success	a user wants to evaluate a potential housemate	they open their profile	a detailed compatibility summary is displayed showing areas of strong match and potential differences
AC-US12-03	Success	a potential housemate views the user's profile	they open it	they receive a notification that someone has viewed their profile
AC-US12-04	Error	a user has not completed their lifestyle preference profile	they attempt to search for compatible housemates	the system prompts them to complete their profile before search results can be personalised
AC-US12-05	Error	a user sets contradictory preferences (e.g. early riser and late sleeper simultaneously)	they attempt to save	the system flags the conflict and prompts them to review their selections
AC-US12-06	Edge case	no housemates match a user's strict preference filters	the search returns zero results	the system suggests broadening one or more filters and displays the closest available matches

Table 19. Acceptance criteria for US-13 (Jira ST4-43).

Acceptance ID	Scenario	Given	When	Then
AC-US13-01	Success	a student enters their university name as a reference point	they search for listings	results are displayed ranked by proximity to the selected campus
AC-US13-02	Success	a student applies filters for maximum weekly rent, distance from campus, number of bedrooms and move-in date	they search	only listings matching all applied filters are displayed
AC-US13-03	Success	a student finds a listing of interest	they open it	they can view affordability indicators including weekly rent in AUD, estimated cost per person, a comparison to the local suburb average and a Budget/Moderate/Premium affordability rating
AC-US13-04	Error	a student searches for listings in an area with no available results	the search returns empty	the system notifies them and suggests broadening their search radius or adjusting filters
AC-US13-05	Error	a listing contains inaccurate photos or descriptions	a student reports it	the listing is flagged for review and temporarily hidden from search results if multiple reports are received
AC-US13-06	Edge case	a student's preferred university is not listed in the system	they search	the system allows them to enter a custom address or suburb as an alternative reference point

6.2 EP-02 Shared Household Living

Table 20. Acceptance criteria for US-01 (Jira ST4-23).

Acceptance ID	Scenario	Given	When	Then
AC-US01-01	Success	a user has created a household and added housemates	they set up a chore roster with task assignments and a rotation schedule	the system distributes chores evenly and notifies each housemate of their assigned tasks
AC-US01-02	Success	a housemate has been assigned a chore	they mark it as complete	the system logs the completion with a timestamp and updates the participation summary
AC-US01-03	Success	a user wants to review contributions	they access the chore history	they can see a full log of who completed what and when
AC-US01-04	Error	a user attempts to create a roster without assigning all housemates	they submit	the system prompts them to assign at least one task to each housemate before activating
AC-US01-05	Error	a chore deadline has passed without being marked complete	the system checks for overdue tasks	it flags the chore as overdue and notifies the assigned housemate
AC-US01-06	Edge case	a housemate leaves the household	their assigned chores become unassigned	the system alerts the house admin to reassign those tasks

Table 21. Acceptance criteria for US-02 (Jira ST4-24).

Acceptance ID	Scenario	Given	When	Then
AC-US02-01	Success	a housemate has added household rules and expectations	an international student opens the household rules section	they can view all rules in a single clearly organised list
AC-US02-02	Success	a user has set their language preference to a non-English language	they view household rules	the rules are displayed in their preferred language or simplified English
AC-US02-03	Success	a user wants clarification on a specific rule	they tap on it	they can view an expanded description or explanation provided by the housemate who created it
AC-US02-04	Error	a user attempts to save a rule without a title or description	they submit	the system prompts them to complete all required fields before saving
AC-US02-05	Error	no household rules have been added yet	a user opens the rules section	the system displays a prompt encouraging housemates to add expectations
AC-US02-06	Edge case	a rule is written in English but the user's preferred language is different	the translation service is unavailable	the system displays the rule in simplified English as a fallback

Table 22. Acceptance criteria for US-03 (Jira ST4-26).

Acceptance ID	Scenario	Given	When	Then
AC-US03-01	Success	a user wants to establish a chore system	they submit a proposed chore system including task assignments and schedules	all housemates receive a notification to review the proposal
AC-US03-02	Success	all housemates have reviewed the proposal	they all approve it	the chore system becomes active and assignments are distributed accordingly
AC-US03-03	Success	a housemate wants to suggest changes	they submit feedback on the proposal	the proposer receives a notification and can revise and resubmit
AC-US03-04	Error	a housemate does not respond to the proposal within 7 days	the deadline passes	the system sends a reminder and flags the proposal as awaiting response
AC-US03-05	Error	a user submits a proposal without assigning tasks to all housemates	they attempt to submit	the system prompts them to assign at least one task per housemate
AC-US03-06	Edge case	one housemate rejects the proposal	the rejection is submitted	the system notifies the proposer and pauses activation until a revised proposal is approved by all

Table 23. Acceptance criteria for US-04 (Jira ST4-28).

Acceptance ID	Scenario	Given	When	Then
AC-US04-01	Success	a user has paid for a shared expense	they log it with a description, amount, category and payer	the system automatically calculates each selected housemate's share and notifies them
AC-US04-02	Success	an expense has been logged	a housemate views their dashboard	they can see all expenses they are involved in and their individual amount owed
AC-US04-03	Success	a housemate has paid their share	the payer marks it as settled	the balance between those two housemates updates automatically
AC-US04-04	Error	a user submits an expense without entering an amount	they attempt to save	the system prompts them to complete all required fields
AC-US04-05	Error	a user attempts to split an expense among housemates not in their household	they submit	the system only allows splitting among confirmed household members
AC-US04-06	Edge case	a housemate leaves the household with an outstanding balance	their account is removed	the system retains their balance record and flags it as unresolved for the remaining housemates

Table 24. Acceptance criteria for US-05 (Jira ST4-29).

Acceptance ID	Scenario	Given	When	Then
AC-US05-01	Success	expenses have been logged between housemates	a user opens their balance dashboard	they can see a clear summary of who owes them money and how much
AC-US05-02	Success	a housemate has paid their share	the user marks the balance as settled	the dashboard updates automatically to reflect the new balance
AC-US05-03	Success	balances exist across multiple housemates	the user views the dashboard	all balances are displayed individually per housemate
AC-US05-04	Error	a user attempts to mark a balance as settled that has already been settled	they submit	the system notifies them that the balance is already cleared
AC-US05-05	Error	no expenses have been logged yet	a user opens the balance dashboard	the system displays a prompt to add the first shared expense
AC-US05-06	Edge case	a balance is disputed by a housemate	they flag it	the system marks the balance as disputed and notifies both parties to resolve it before it can be settled

Table 25. Acceptance criteria for US-06 (Jira ST4-32).

Acceptance ID	Scenario	Given	When	Then
AC-US06-01	Success	a utility bill has been received	a user enters the bill type, total amount and billing period	the system calculates and assigns each housemate's share based on the selected split method
AC-US06-02	Success	a bill has been logged	all housemates view the shared bill history	they can see all submitted bills and their individual amounts owed
AC-US06-03	Success	a housemate believes a bill is incorrect	they flag it as disputed	all housemates are notified and the bill is marked as under review
AC-US06-04	Error	a user submits a bill without entering a total amount	they attempt to save	the system prompts them to complete all required fields
AC-US06-05	Error	a user attempts to split a bill among zero housemates	they submit	the system requires at least two participants before allowing submission
AC-US06-06	Edge case	a utility bill covers a period during which a housemate was absent	the bill is split	the system allows the user to manually exclude or adjust that housemate's share for that billing period

Table 26. Acceptance criteria for US-07 (Jira ST4-33).

Acceptance ID	Scenario	Given	When	Then
AC-US07-01	Success	a user wants to post a notice	they create a notice with a title, description and optional image	it is immediately visible to all housemates on the noticeboard
AC-US07-02	Success	a notice has been posted	a housemate wants to respond	they can leave a limited reaction or reply directly on the notice
AC-US07-03	Success	a notice has an expiry date set	the date passes	the notice is automatically archived and removed from the active noticeboard
AC-US07-04	Error	a user attempts to post a notice without a title	they submit	the system prompts them to add a title before posting
AC-US07-05	Error	a user sets an expiry date in the past	they submit	the system rejects the date and prompts them to select a future date
AC-US07-06	Edge case	a notice receives no interaction for 30 days and has no expiry date set	the system checks for inactive notices	it prompts the creator to either renew or archive the notice

Table 27. Acceptance criteria for US-08 (Jira ST4-34).

Acceptance ID	Scenario	Given	When	Then
AC-US08-01	Success	a user wants to raise an issue	they submit it with the anonymous option selected	the issue is posted to the shared household space without revealing the submitter's name or profile
AC-US08-02	Success	an issue has been posted	housemates view it	they can vote on whether they agree it is a concern
AC-US08-03	Success	all parties have agreed the issue is resolved	the submitter or a majority vote marks it as resolved	the issue is closed and archived
AC-US08-04	Error	a user attempts to submit an issue without a description	they submit	the system prompts them to add a description before posting
AC-US08-05	Error	a user attempts to identify an anonymous submitter through the system	they try to access submitter information	the system does not reveal any identifying information under any circumstance
AC-US08-06	Edge case	an anonymous issue receives no votes or responses within 14 days	the system checks for inactive issues	it sends a reminder to all housemates to review and respond

Table 28. Acceptance criteria for US-10 (Jira ST4-36).

Acceptance ID	Scenario	Given	When	Then
AC-US10-01	Success	a user wants to propose a house rule	they submit it with a title, description and category	all housemates receive a notification to review and acknowledge it
AC-US10-02	Success	all housemates have acknowledged a proposed rule	the last acknowledgement is submitted	the rule is officially added to the household rules list with a timestamp
AC-US10-03	Success	a user wants to reference a rule during a dispute	they link it to a household issue	the rule is displayed alongside the issue for all housemates to see
AC-US10-04	Error	a user attempts to submit a rule without a title or category	they submit	the system prompts them to complete all required fields
AC-US10-05	Error	a housemate attempts to edit or remove a rule without majority approval	they submit the change	the system rejects it and notifies all housemates that a change has been requested pending approval
AC-US10-06	Edge case	a rule has not been acknowledged by one housemate after 7 days	the system checks for pending acknowledgements	it sends a reminder to that housemate to review and acknowledge the rule

6.3 EP-03 Accountability Extensions

Table 29. Acceptance criteria for US-14 (Jira ST4-25).

Acceptance ID	Scenario	Given	When	Then
AC-US14-01	Success	a chore deadline has passed and the chore has not been marked complete	the system checks for overdue tasks	it automatically sends an in-app notification to the assigned housemate
AC-US14-02	Success	the house admin wants to manually remind a housemate	they send a nudge	the assigned housemate receives an in-app notification without any personal contact details being exchanged
AC-US14-03	Success	a housemate marks an overdue chore as complete after receiving a reminder	they submit the completion	the system updates the chore log and removes the overdue flag
AC-US14-04	Error	a housemate has disabled notifications	a chore becomes overdue	the system still flags the chore as overdue within the app and displays it prominently on their dashboard
AC-US14-05	Edge case	a housemate is temporarily removed from the roster (e.g. on leave)	their chore deadline passes	the system does not send them an overdue reminder and reassigns the task to the next available housemate

Table 30. Acceptance criteria for US-15 (Jira ST4-41).

Acceptance ID	Scenario	Given	When	Then
AC-US15-01	Success	a confirmed tenancy period has ended	a user submits a written review and star rating for their previous housemate	the review is displayed on that housemate's profile in chronological order
AC-US15-02	Success	a user views a housemate's profile	they scroll to the reviews section	they can see all reviews with the overall star rating displayed prominently at the top
AC-US15-03	Success	a housemate receives a review	it is posted	they receive a notification and can submit a single written response
AC-US15-04	Error	a user attempts to submit a review before a confirmed tenancy period has ended	they try to submit	the system prevents submission and notifies them that reviews can only be left after a confirmed tenancy
AC-US15-05	Error	a user attempts to submit a second review for the same housemate from the same tenancy	they submit	the system rejects the duplicate and notifies the user
AC-US15-06	Edge case	a review is flagged as inappropriate by multiple users	the system detects multiple reports	the review is temporarily hidden pending platform review and the reviewer is notified

6.4 EP-04 Property Management

Table 31. Acceptance criteria for US-16 (Jira ST4-30).

Acceptance ID	Scenario	Given	When	Then
AC-US16-01	Success	a property manager has set a rent amount and due date for a tenant	the due date approaches	the system automatically sends a rent reminder to that tenant
AC-US16-02	Success	a tenant's payment has been recorded	the property manager views the dashboard	the tenant's status updates to "paid"
AC-US16-03	Success	the due date has passed without payment being recorded	the system checks payment status	the tenant's status updates to "overdue" and the property manager is notified
AC-US16-04	Error	a property manager attempts to set a rent due date without entering a rent amount	they submit	the system prompts them to complete all required fields
AC-US16-05	Error	a tenant's contact details are incomplete	the system attempts to send a reminder	the system flags the tenant's profile as incomplete and alerts the property manager
AC-US16-06	Edge case	a tenant disputes a rent amount	they flag it through the app	the system marks the payment status as disputed and notifies the property manager to review

Table 32. Acceptance criteria for US-17 (Jira ST4-40).

Acceptance ID	Scenario	Given	When	Then
AC-US17-01	Success	a property manager has completed a listing with photos, description, rent price, available date and house rules	they publish it	it immediately appears in student search results based on location and filters
AC-US17-02	Success	a student finds a listing they are interested in	they send an enquiry	the property manager receives a notification and the message is stored in their centralised inbox
AC-US17-03	Success	a listing's available date has passed	the system checks active listings	the listing is automatically removed from search results and the property manager is notified
AC-US17-04	Error	a property manager attempts to publish a listing without photos or a rent price	they submit	the system prompts them to complete all required fields before publishing
AC-US17-05	Error	a student attempts to enquire about a listing that has already been filled	they submit the enquiry	the system notifies them that the listing is no longer available
AC-US17-06	Edge case	a listing has been live for 60 days without being filled	the system checks listing age	it prompts the property manager to review and update the listing details

Table 33. Acceptance criteria for US-18 (Jira ST4-37).

Acceptance ID	Scenario	Given	When	Then
AC-US18-01	Success	a property manager wants to send an announcement	they compose it with a title, description, category and optional attachment	it is delivered to all tenants in the selected property
AC-US18-02	Success	an announcement has been sent	a tenant opens the app	they receive a push notification and can view the announcement in their notification centre
AC-US18-03	Success	a property manager wants to schedule a recurring announcement	they set a recurrence schedule	the system automatically sends the announcement at the defined intervals
AC-US18-04	Error	a property manager attempts to send an announcement without selecting a property	they submit	the system prompts them to select at least one property before sending
AC-US18-05	Error	a scheduled announcement fails to send	the system detects the failure	it retries delivery and notifies the property manager if the issue persists
AC-US18-06	Edge case	a tenant has not viewed a high priority announcement within 24 hours	the system checks read receipts	it sends a follow-up push notification to that tenant

7 Functional and Non-Functional Requirements

7.1 Functional Requirements

Required content

Define what the system must do. Link every requirement to one or more user stories and state how the requirement will be verified.

Table 34. Functional requirements template.

Requirement ID	Functional requirement	Related stories	Verification method
FR-01	The system shall <i>[required behaviour]</i> .	<i>[US IDs]</i>	<i>[Test or prototype evidence]</i>
FR-02	The system shall <i>[required behaviour]</i> .	<i>[US IDs]</i>	<i>[Test or prototype evidence]</i>
FR-03	The system shall <i>[required behaviour]</i> .	<i>[US IDs]</i>	<i>[Test or prototype evidence]</i>

7.2 Non-Functional Requirements

Required content

Include measurable requirements for privacy, security, accessibility, performance and usability. Add other categories where they are relevant to the solution.

Table 35. Non-functional requirements template.

Requirement ID	Category	Measurable requirement	Verification method
NFR-01	Privacy	<i>[Privacy target]</i>	<i>[Verification method]</i>
NFR-02	Security	<i>[Security target]</i>	<i>[Verification method]</i>
NFR-03	Accessibility	<i>[Accessibility target]</i>	<i>[Verification method]</i>
NFR-04	Performance	<i>[Performance target]</i>	<i>[Verification method]</i>
NFR-05	Usability	<i>[Usability target]</i>	<i>[Verification method]</i>

8 User Story Dependencies

Each user story maps to one goal level use case. The tables use those use case identifiers and the mapping in Appendix A. Supporting prerequisites are included when no user story creates the required data.

8.1 Dependency Types

Table 36. Dependency types.

Type	Meaning
Required	The dependent use case cannot proceed until the prerequisite is satisfied.
Required data	The dependent use case needs data created by the prerequisite.
Required data OR	Any one listed source can provide the data.
Conditional	The prerequisite applies only under the stated condition.
Optional	The prerequisite improves the path but does not block it.

8.2 Dependency Tables

Table 37. Dependencies within EP-02.

Dependent use case	Depends on	Type	Reason
UC-01 Chore roster	SUP-02 Household membership	Required	Only current household members can manage the roster.
UC-01 Chore roster	UC-03 Approved chore system	Conditional	The approved structure is required when the household uses the approval workflow.
UC-02 View household rules	SUP-02 Household membership	Required	Rules belong to one household.
UC-02 View household rules	UC-10 Agreed house rules	Required data	At least one acknowledged rule must exist before it can be displayed.
UC-03 Propose chore system	SUP-02 Household membership	Required	The system must identify every person who can review the proposal.
UC-04 Log shared expenses	SUP-02 Household membership	Required	The expense and its shares belong to one household.
UC-05 Track balances	UC-04 Logged expenses	Required data	Balances are calculated from recorded expenses.
UC-06 Split utility bills	UC-04 Expense calculation	Required	Utility bills reuse the shared expense calculation.
UC-07 Household notices	SUP-02 Household membership	Required	Only current members can post or view notices.
UC-08 Anonymous issues	SUP-02 Household membership	Required	The system must confirm household access while hiding the reporter from other residents.
UC-10 Agreed house rules	SUP-02 Household membership	Required	Only current members can propose or acknowledge a rule.

Table 38. Dependencies within EP-01.

Dependent use case	Depends on	Type	Reason
UC-09 Secure messaging	UC-11 Verification status	Optional	Messaging works without verification. The status adds a caution signal.
UC-11 View verification status	SUP-04 Verification record or DATA-02 Seeded profiles	Required data OR	A stored status must exist before the interface can display it.
UC-12 Compare lifestyle fit	DATA-02 Seeded profiles	Required data	The preliminary prototype needs completed sample profiles for comparison.
UC-13 Search listings	UC-17 Published listings or DATA-01 Seeded listings	Required data OR	At least one published listing must exist before search can return a result.

Table 39. Dependencies within EP-03 and EP-04.

Dependent use case	Depends on	Type	Reason
UC-14 Overdue reminders	UC-01 Chore roster or UC-04 Shared expenses	Required data OR	A reminder needs an overdue assignment or payment.
UC-14 Overdue reminders	SUP-03 Notifications	Required	The service delivers the reminder.
UC-15 Read housemate reviews	SUP-02 Former household or tenancy record	Required data	The system must confirm that each displayed review came from a previous shared living arrangement.
UC-16 Rent tracking	SUP-02 Tenancy membership	Required	Rent status belongs to a confirmed tenant and property.
UC-16 Rent tracking	SUP-03 Notifications	Optional	Notifications can deliver a reminder outside the active app.
UC-17 Manage listings	SUP-04 Verification record	Conditional	A verification record is required only when the interface shows a verified status.
UC-18 Tenant announcements	SUP-02 Tenancy membership	Required	The system needs the current tenant audience for the selected property.
UC-18 Tenant announcements	SUP-03 Notifications	Optional	Notifications can deliver the update outside the active app.

8.3 Dependency Graph

Figure 3 visualises the same relationships. Solid arrows show required dependencies. Dotted arrows show optional or conditional support. Diamond nodes show an OR condition.

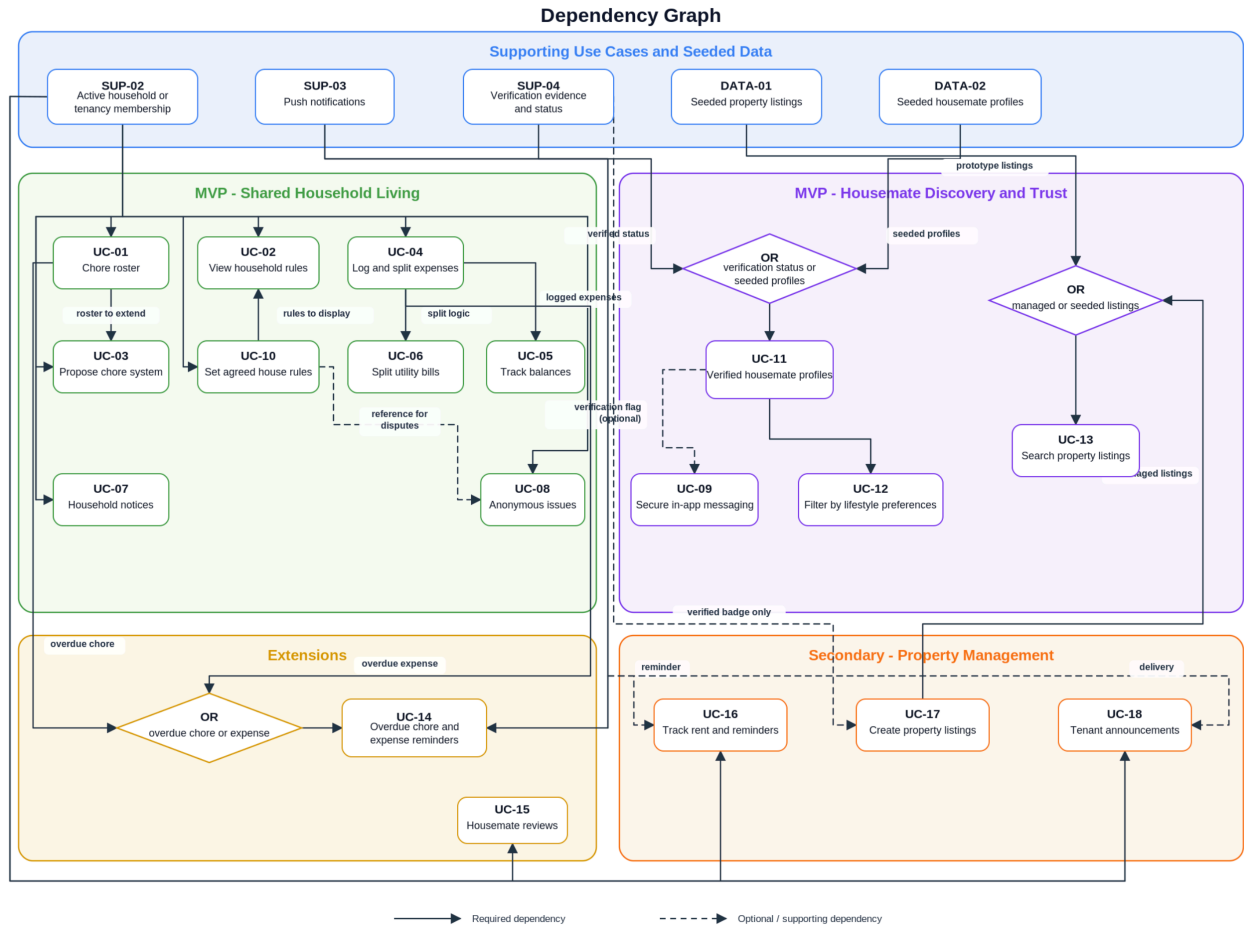


Figure 3. Use case dependencies for the product backlog.

9 First-Pass User Story Map

Required content

Show the user activities, user tasks and release slices. Clearly identify which stories belong to Sprint 1 and which stories are scheduled for later sprints or the backlog.

Table 40. First-pass user story map template.

Release slice	[Activity 1]	[Activity 2]	[Activity 3]	[Activity 4]	[Activity 5]
User task	[Task]	[Task]	[Task]	[Task]	[Task]
Sprint 1	[US IDs]	[US IDs]	[US IDs]	[US IDs]	[US IDs]
Later sprint	[US IDs]	[US IDs]	[US IDs]	[US IDs]	[US IDs]
Backlog	[US IDs]	[US IDs]	[US IDs]	[US IDs]	[US IDs]

9.1 Sprint 1 Scope and Rationale

Required content

List the Sprint 1 stories and explain why they form a coherent, testable path through the selected concept.

[Insert the approved Sprint 1 story IDs and rationale.]

10 Preliminary Prototype

10.1 Prototype Scope

Required content

The preliminary prototype must correspond to the selected user stories. It must demonstrate the main Sprint 1 path and allow the related features to be tested.

[Describe the prototype scope, target users and main task flow.]

10.2 Prototype Flow

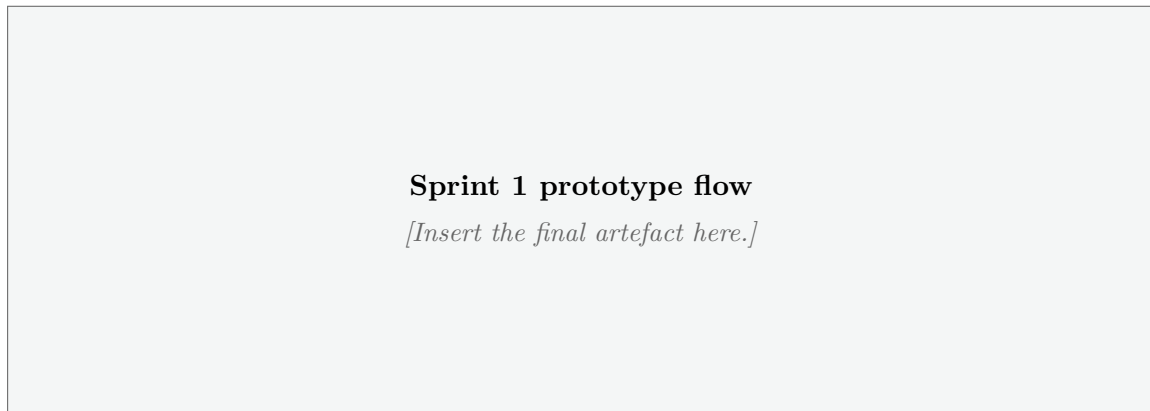


Figure 4. Sprint 1 prototype flow.

10.3 Prototype Screens

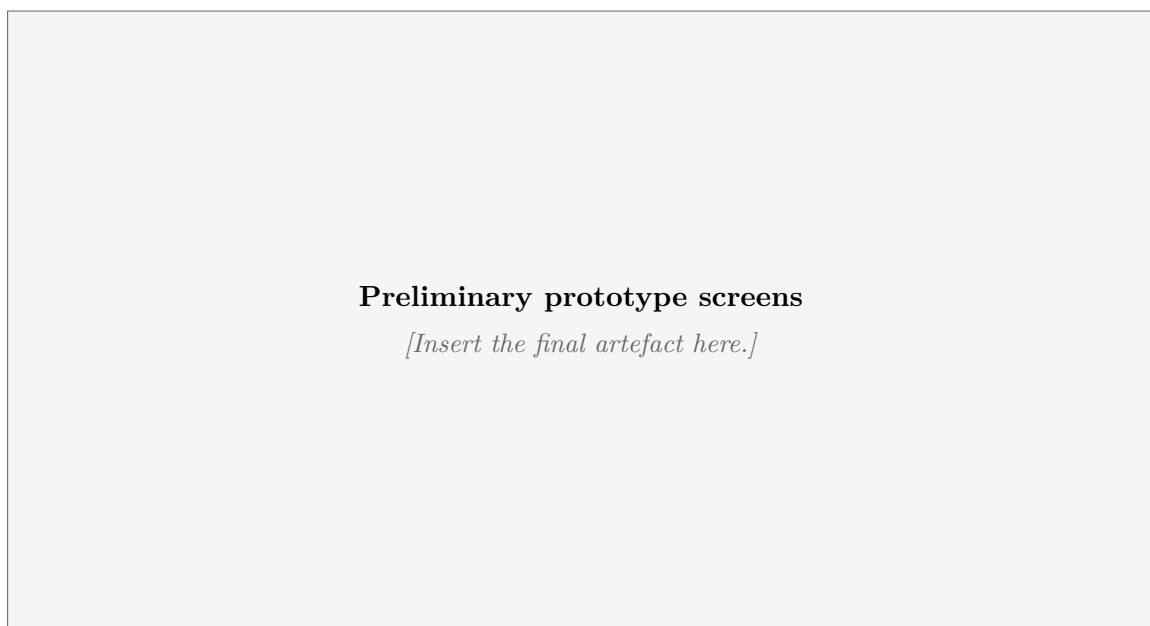


Figure 5. Preliminary prototype screens.

11 Prototype-to-User-Story Traceability

Each screen of the HTML prototype is mapped to the user story and the acceptance criteria that it demonstrates, so every Sprint 1 requirement can be traced from the backlog to an interaction that can be exercised in the prototype. Table 41 records this mapping, and the same screen identifiers are used by the test plan in Section 12. Together the two sections close the chain that starts at a research finding and ends at a testable screen. The link column is completed once the prototype pages are hosted.

Table 41. Prototype-to-user-story traceability.

Screen ID	Prototype screen	Story ID	Acceptance ID	Testable interaction	Link
S-01	Lifestyle profile setup	US-12	AC-US12-04, AC-US12-05	Complete the required preference fields, submit an incomplete form to trigger the completion prompt, save contradictory selections to trigger the conflict flag.	
S-02	Housemate and listing search results	US-13, US-11	AC-US13-01, AC-US13-02, AC-US11-02	Search from a university reference point, apply rent and distance filters, switch on the verified-only filter.	
S-03	Listing and profile detail	US-11, US-13	AC-US11-03, AC-US13-03	Open a verified and an unverified profile to compare the badge with the warning indicator, locate the affordability details.	
S-04	Compatibility comparison	US-12	AC-US12-01, AC-US12-02, AC-US12-06	Compare two candidate profiles through the compatibility summary, apply a strict filter to reach the zero-result broadening suggestion.	
S-05	Guided in-app chat	US-09	AC-US09-01, AC-US09-04, AC-US09-06	Send a first message from a guided prompt, type contact details to trigger the warning, view the unverified sender notice.	
S-06	House rule proposal	US-10	AC-US10-01, AC-US10-04	Propose a rule with a title, description and category, submit with missing fields to trigger the completion prompt.	
S-07	Household rules list and acknowledgement	US-10, US-02	AC-US10-02, AC-US02-01	Acknowledge a proposed rule as each housemate, then find the active rule in the rules list.	

12 Prototype Test Plan

Five moderated think aloud tasks will assess the Sprint 1 path. Each task uses five participants from the target group. For each task, record whether the user finishes, how long it takes, where errors occur, and which acceptance criterion fails.

Table 42. Prototype test plan.

Test	Stories	Participant	Task	Pass condition	Owner
T1	US-12	Student new to shared housing	Complete the lifestyle profile. Try one incomplete submission and one contradictory selection.	At least 4 of 5 users finish within 3 minutes without help. The prototype identifies both invalid states.	Pranav
T2	US-11, US-13	Student searching for a room	Find a room under \$350 per week. Identify the listing status and the profile status. Open the property details.	At least 4 of 5 users report both statuses and find the required details within 2 minutes. The unverified warning is understood.	Nicky
T3	US-12	Student comparing housemates	Compare two profiles and choose the better match. Apply a filter that produces no result.	At least 4 of 5 users choose the higher score and state one mismatch. The broadening prompt is noticed.	Liam
T4	US-09	International student making first contact	Contact a listing with a guided prompt. Attempt to enter personal contact details.	At least 4 of 5 users send a message within 2 minutes. A warning appears before contact details are shared. The unverified notice is understood.	Dylan
T5	US-02, US-10	Student moving into a new household	Propose a quiet hours rule. Complete the acknowledgements. Find the active rule.	At least 4 of 5 users finish within 3 minutes. An incomplete proposal is rejected and the active state is clear.	Liam

After each round, the team records failed criteria and revises the affected screens. The next round retests those criteria.

13 Conclusion

Section purpose

Summarise the selected concept, the Sprint 1 scope, the preliminary prototype and the next validation step. Do not introduce new evidence in the conclusion.

[Insert the conclusion after the report content is complete.]

14 References

- *[Insert references reused from Part 1 and any new sources used in Part 2.]*

A Full User Story and Use Case Mapping

Table 43 links each user story to one goal level use case. The scope values match Table 10.

Table 43. User story and use case mapping.

Story	Epic	Use case	Primary actor	Successful outcome	Scope
US-01	EP-02	UC-01 Manage a chore roster	Share house resident	The user assigns rotating chores and can view completion history.	Later sprint
US-02	EP-02	UC-02 View household rules	Share house resident	The user finds the accepted rules for the current household.	Sprint 1
US-03	EP-02	UC-03 Propose and approve a chore system	Share house resident	The proposed system becomes active only after the required approvals.	Later sprint
US-04	EP-02	UC-04 Log a shared expense	Share house resident	The user records the payer and the system calculates each selected share.	Later sprint
US-05	EP-02	UC-05 Track balances	Share house resident	The user sees current balances derived from recorded expenses.	Later sprint
US-06	EP-02	UC-06 Split a utility bill	Share house resident	The user divides a utility bill among selected household members.	Later sprint
US-07	EP-02	UC-07 Manage household notices	Share house resident	The user posts or reads a notice within the household.	Backlog
US-08	EP-02	UC-08 Raise an anonymous issue	Share house resident	Other residents can view the issue without seeing the reporter's identity.	Backlog
US-09	EP-01	UC-09 Send secure messages	Accommodation seeker	The user sends and receives messages without exposing personal contact details.	Sprint 1
US-10	EP-02	UC-10 Propose and acknowledge a house rule	Share house resident	A proposed rule becomes active after the required acknowledgements.	Sprint 1
US-11	EP-01	UC-11 View verification status	Accommodation seeker	The user can distinguish verified and unverified profiles.	Sprint 1
US-12	EP-01	UC-12 Compare lifestyle fit	Accommodation seeker	The user sees a compatibility result with the factors that affected it.	Sprint 1
US-13	EP-01	UC-13 Search property listings	Accommodation seeker	The user applies location and cost filters to the available listings.	Sprint 1
US-14	EP-03	UC-14 Receive overdue reminders	Share house resident	The user receives a reminder for an overdue chore or payment.	Later sprint
US-15	EP-03	UC-15 Read housemate reviews	Accommodation seeker	The user reads reviews linked to confirmed former arrangements.	Backlog
US-16	EP-04	UC-16 Track rent status	Property manager	The manager records rent status and sends an overdue reminder.	Secondary
US-17	EP-04	UC-17 Manage property listings	Property manager	A complete listing is published and becomes searchable.	Secondary
US-18	EP-04	UC-18 Send tenant updates	Property manager	An update reaches the current tenants of the selected property.	Secondary

Supporting use cases serve several user goals. Seeded data allows the preliminary prototype to demonstrate discovery without the property manager workflow.

Table 44. Supporting use cases.

ID	Supporting use case	Primary actor	Successful outcome
SUP-01	Manage account and role access	Registered user	The user is authenticated and receives the correct permissions.
SUP-02	Establish household or tenancy membership	Resident or property manager	Membership becomes active only after confirmation.
SUP-03	Deliver notifications	System	A reminder or update can reach a user outside the active app.
SUP-04	Manage verification evidence and status	User or system	Evidence is stored and the related status is available to the interface.

Table 45. Supporting prototype data.

Data ID	Data source	Purpose
DATA-01	Seeded property listings	Supplies published listings before the property manager workflow is implemented.
DATA-02	Seeded housemate profiles	Supplies completed profiles for verification and compatibility testing.

B Prototype Screens and Links

Appendix content

Insert full-size prototype screenshots, hosted page links and any interaction notes needed for assessment.

C Jira Evidence

Appendix content

Insert Jira backlog exports, screenshots or links showing Epics, user stories, acceptance criteria and sprint assignment.